MIL-T-5913D(USAF)
15 February 1988
SUPERSEDING
MIL-T-5913C(USAF)
13 January 1975

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, SYNCHRO TYPE J-5

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating synchro fuel flow-rate transmitter, range 100 to 2,000 pph, designated Type J-5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Wood, Boxes Cleated Plywood
PPP-B-636	Box, Shipping Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
MIL-T-5350	Transmitter, Synchro Operated, Aircraft, General Specification For
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-I-5914	Indicator, Rate Of Flow, Fuel, Type A-16

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-2073-1	DOD Material Procedures For Development And Application Packaging Requirements
MS33649	Bosses, Fluid Connection-Internal Straight Thread

(Copies of specifications, standards, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and which has been listed on or approved for listing on the applicable Qualified Products List (4.3 and 6.3).

3.2 General specification. The requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

3.3.1 Design. Unless otherwise specified in the contract or order, the fuel flow transmitter shall be designed to transmit from a remote station, the rate of flow in pounds per hour (pph) of 100/130 grade aviation gasoline with a specific gravity of 0.715 ± 0.003 between 15° and 25°C, and in accordance with MIL-G-5572.

3.3.2 Operation. The calibration of the transmitter shall be such that it will satisfactorily operate a Type A-16 indicator conforming to MIL-I-5914.

3.3.3 Calibration. Each transmitter shall be calibrated and adjusted for maximum accuracy at the rate of flow specified in Table I as the "setting point".

3.3.4 Construction. The transmitter body screws shall be safetied with stainless steel lockwire. Not more than three screws shall be safetied with one length of wire. The wire shall be twisted tight between each pair of screws. Suitable lockwashers may be used in place of lockwire, where necessary.

3.4 Rust inhibitor. A rust inhibitor of a quality and type acceptable to the contracting activity shall be placed in each transmitter prior to sealing for shipment.

3.5 Dimensions. The outline dimensions of the transmitter shall conform to Figure 1. Connection bosses shall conform to MS33649 for 1.250 inch outside diameter tubing, except that the "G" diameter shall be a minimum of 2.062-inches.

TABLE I. Fuel Flowmeter Transmitter.

Test Point (pph)	Tolerance (pph)			
	Room Temperature	High Temperature	Low Temperature	Substitute Fuel
100	15	40	30	20
200	7	25	20	10
400 Setting Point	5	20	15	10
600	7	20	15	15
800	12	25	25	20
1,000	20	35	40	30
1,200	25	40	60	45
1,400	30	50	75	55
1,600	30	55	90	65
1,800	35	60	100	70
2,000	35	65	110	80

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

3.6 Weight. The weight of the complete fuel flowmeter transmitter shall not exceed 3.5 pound.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8 Warning instruction. With each transmitter, one printed or reproduced copy of the following warning instructions shall be included in a prominent manner: "USE 1.625 12N-3 STRAIGHT THREAD FITTINGS ONLY TO CONNECT TRANSMITTER. DO NOT USE PIPE THREAD FITTINGS."

3.9 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Qualification inspection. Qualification inspection shall consist of performing all the examinations and tests of this specification. Test samples shall consist of three transmitters unless otherwise specified by the qualifying activity, tests shall be performed in the order listed.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of individual tests and sampling plans and tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

4.4.1 Individual tests. Each transmitter shall be subjected to the individual tests of MIL-T-5350. In addition, the sealing test (4.6.2) and the leak test (4.6.3) herein, shall be conducted.

4.4.2 Sampling plans and tests.

4.4.2.1 Sampling Plan A. Sampling Plan A tests shall be in accordance with MIL-T-5350.

4.4.2.2 Sampling Plan A.

4.4.2.2.1 Omit. The following tests listed under sampling Plan B of MIL-T-5350 shall be omitted.

(a) Seasoning
(b) Damping

4.4.2.2.2 Add. In addition to the tests listed under sampling Plan B of MIL-T-5350 (except those specified in 4.4.2.2.1 herein) the following tests shall be conducted:

(a)	Pressure drop	(4.6.5)
(b)	Substitute fuel	(4.6.9)
(c)	Bypass	(4.6.10)

4.5 Test conditions. Unless otherwise specified, the transmitter shall be tested against an indicating standard constructed as specified in MIL-T-5350. The transmitter shall be tested with the center line of the self-synchronous motor in a horizontal plane which is below the center line of the fuel hose.

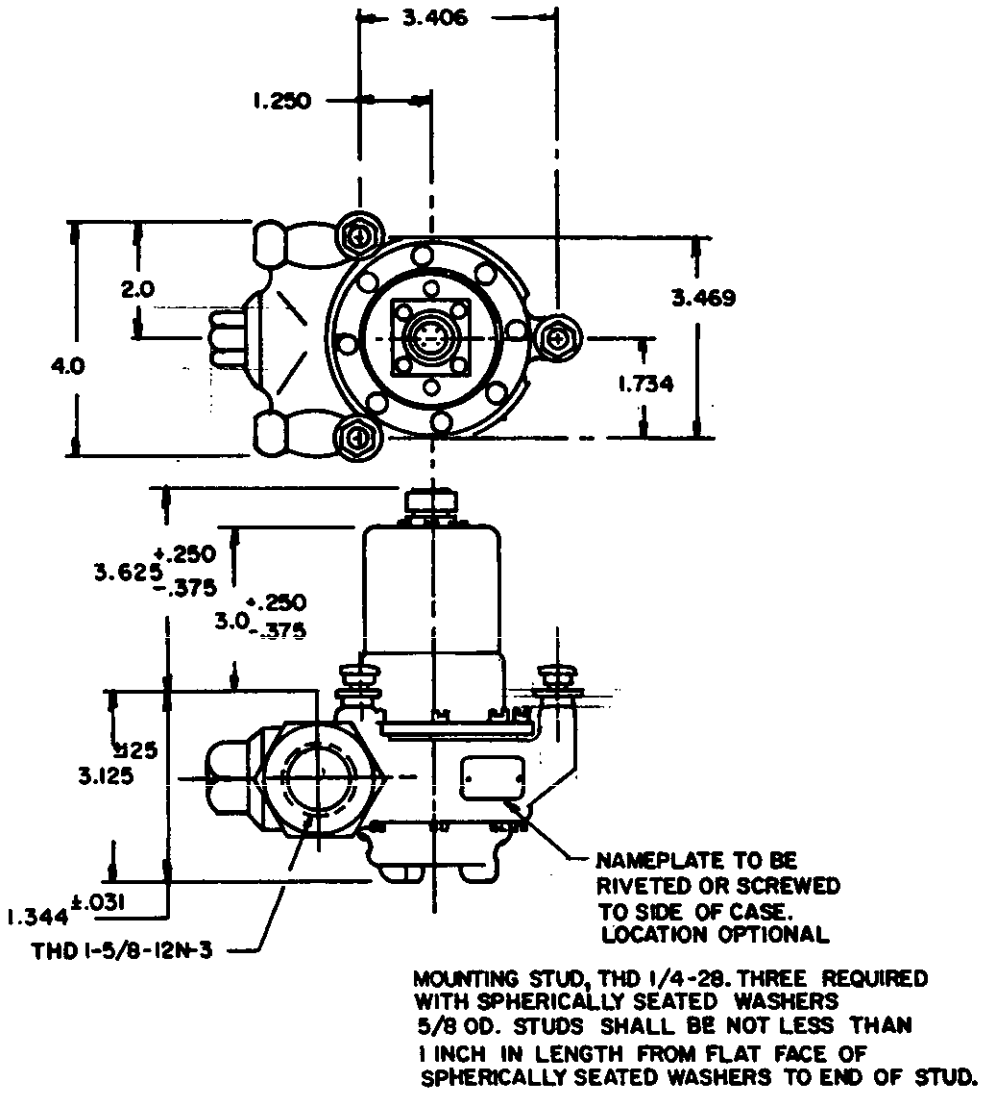
4.6 Test methods.

4.6.1 Scale error and friction. The transmitter shall be tested for scale errors and friction at the points shown in Table I. The errors at the test points shall not exceed the tolerances specified in Tables I and II.

TABLE II. Fuel flowmeter Transmitter.

Test	Tolerance (pph)
Friction	
Room Temperature	10
High Temperature	10
Low Temperature	15
Vibration	
Pointer oscillation	20
Pointer variation	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES DECIMALS ±.005.

FIGURE 1. Transmitter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

4.6.2 Sealing. Each transmitter, with fuel line caps and electrical connections installed, shall be subjected to an internal case pressure of 14 pounds per square inch (psi) for a minimum period of 30 seconds. The entire transmitter, with pressure applied, shall be immersed in a suitable liquid, such as water. Any bubbles coming from within the transmitter case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various external parts of the case shall not be considered as leakage.

4.6.3 Leaks. The transmitter shall be tested for leaks immediately after the scale error test while the flow chamber is still moist from fuel, but with all free fuel drained out. An air pressure of 40 psi shall be applied to the flow chamber. With the air supply disconnected, no change shall be observed in the pressure over a period of 5 minutes. The pressure gage used shall be such that a pressure change of 0.125 psi can be easily detected while the gage is being tapped. The piping from the air supply to the gage shall be as short as practicable, and shall have an inside diameter not to exceed 0.250 inch.

4.6.4 Position error. Readings of the master indicator shall be determined with the transmitter in the positions specified in MIL-T-5350, and shall not differ from readings taken while the transmitter is in normal position by more than 10 pph.

4.6.5 Pressure drop. The pressure drop within the transmitter for rates of flow specified shall be determined, and shall not exceed the tolerances specified in Table III, for normal

TABLE III. Fuel Flowmeter Transmitter.

Flow rate (pph)	Maximum allowable Pressure drop (psi) Normal	Blocked vane
2,000	1.25	3.250
3,000	2	3,500

4.6.6 Vibration. The transmitter shall be subjected to vibration for a total of 3 hours on a suitably designed vibration stand at a total excursion of not less than 0.060 inch nor more than 0.070 inch. The vibration shall be such that the transmitter motion shall be sinusoidal in a plane inclined at 45 degees to the plane of the mounting studs. The frequency of vibration shall be 2,000 cpm. Following the vibration period, the transmitter shall be tested for scale error and friction as specified in 4.6.1.

4.6.6.1 Pointer oscillation. The transmitter shall be subjected to the test specified in 4.6.6. The frequency of vibration shall be varied gradually from 300 to 3,000 cpm while the fuel flow is slowly varied through the scale range to determine the magnitude of pointer oscillation and variation on a standard single synchro indicator connected electrically to the transmitter. The magnitude of pointer oscillation and variation shall not exceed the tolerance specified in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

4.6.7 High temperature. The transmitter shall be subjected to an ambient temperature of +70° ± 2°C for a period of 4 hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38° ± 2°C shall be pumped through the transmitter at a rate of 500 pph. The transmitter shall then be tested for scale error and friction with the ambient temperature maintained at +70° ± 2°C. The scale error and friction shall not exceed the tolerances specified in Tables I and II.

4.6.8 High temperature exposure. The transitter shall be subjected to ambient temperature of + 70° ± 2°C for a period of 50 hours with fuel in the instrument, but not flowing. At the end of this period and with the transmitter still at this temperature, the transmitter shall be tested for scale error and friction. The errors shall not exceed the errors specified in Table I. During the last 30 minutes of this period, fuel at 38° ± 2°C shall be pumped through the transmitter at a rate of 500 pph.

4.6.9 Substitute fuel. The transmitter shall be subjected to the scale error test, at room temperature, except that the fuel used shall contain 35 to 40 percent aromatics (benzene, xylene and toluene), and specific gravity shall be 0.750 ± 0.01 at room temperature. The scale error shall not exceed the tolerances specified in Table I.

4.6.10 Bypass. The pressure drop, with the transmitting moveable element blocked in the zero flow position, shall be measured by any suitable method and shall not exceed the value specified in Table III, for blocked vane.

4.6.11 Endurance. The scale errors after the endurance test shall not exceed the tolerances specified in Table I for the room temperature scale error test by more than 10 pph. The rate of cycling shall 15 ± 5 cpm.

4.7 Inspection and test. Tests of methods of preservation shall be accomplished in accordance with Section 4 of MIL-P-116, to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quanitity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure transmitter does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type J-5 fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pph being consumed by an aircraft engine of approximately 1,800 to 2,500 takeoff horsepower, in conjunction with a suitable indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Ordering data shall be as specified in MIL-T-5350.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacture are urged to arrange to have the products that they propose to offer to the Federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center ATTN: MMEDO, Tinker Air Force Base, Oklahoma 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word listing).
Gasoline, Grades 80/87, 100/130 and 115/145
Indicator, Type A-16
Synchro Style
C123J and T37
100 to 2,000 PPH

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Reviewer activity:
AIR FORCE -11

Project number:
6620-F455

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

☆ U.S. Government Printing Office: 1982—360-879/5804

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U.S. AIR FORCE

OC-ALC/MMEDO
Tinker AFB OK 73145-5990

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T05:56Z
Check the source to verify that this is the current version before use.